

Mayukh Sinha

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EDUCATION

Northeastern University, College of Engineering, Boston

Expected May 2026

Masters of Science in Software Engineering Systems, **GPA: 4.0**

Relevant Coursework: System Design, Network Structures and Cloud Computing, Web Design and User Experience Program Structures and Algorithms.

Jaypee Institute Of Information Technology, India

2017-2021

Bachelor of Technology in Electronics and Communication Engineering

Relevant Coursework: Software Development Fundamentals, DBMS, Data Structures and Algorithms, Operating Systems.

SPECIALIZED SKILLS

Programming Languages: C#, Java, Python, C++, Javascript

Frameworks & Technologies: ASP.NET, Entity Framework, Specflow, React JS, Node.js, Bootstrap, Git, Unit Testing

Cloud & DevOps: AWS EC2, VPC, S3, IAM, Linux, Docker, Jenkins, Git Actions, Terraform, Packer

Databases: SQL, MongoDB, Migrations

Networking & Security: TCP/IP, API Security, SSH, OAuth.

WORK EXPERIENCE

Software Development Engineer, SOTI Inc., Gurgaon

Jul 2021 - Aug 2024

- Developed and optimized MobiControl backend services using C# and ASP.NET, improving system performance by 20% through code refactoring, query optimization, and caching, ensuring faster response times and scalability. Integrated structured logging and performance profiling using Serilog, enabling proactive monitoring and issue resolution.
- Integrated third-party services to enhance the MobiControl platform's functionality, including push notifications and real-time device monitoring, contributing to business-impacting features that were highlights of SOTI Sync 2023.
- Collaborated closely with the DevOps team to implement CI/CD pipelines, improving deployment frequency and reducing wait times by 30% by improving test cases selection and bucketing on EC2.
- Trained 100+ freshers** in a structured .NET and cloud technologies like **AWS services**, focusing on backend development, API design, and cloud deployment, equipping them with essential skills for project contributions.
- Led a team to implement an in-house device farm to automate and test MC agent builds every day, based on my original idea, **saving \$150,000 every quarter** and receiving the **MVP award in 2023** for this initiative.
- Developed an upgrade framework for MobiControl to ensure reliability in on-premises software updates. Integrated checksum verification to detect and prevent data corruption during update deployment.

RELEVANT ACADEMIC PROJECTS

Cloud Infra, Northeastern University, Boston

Jan 2025 - Ongoing

- Designed a dual-repository structure, segregating infrastructure and application code while enforcing automated reviews and quality checks; incorporated CloudWatch alarms and load balancer health metrics for enhanced visibility.
- Built a CI/CD pipeline to run integration tests on PR creation and generate deployment artifacts for AWS and GCP using Packer automatically registering new instances with the load balancer and displaying metrics on CloudWatch dashboards.
- Automated infrastructure provisioning with Terraform, deploying EC2 instances from AMIs generated by repository merges and configuring load balancer settings for balanced traffic distribution.
- Created an automated deployment validation system that verifies successful rollouts through the API endpoint, with CloudWatch monitoring to track performance and errors, ensuring optimal load balancer integration.

NUConnect, Northeastern University, Boston - [ConnectRoom](#)

Sept 2024 - Dec 2024

- Developed Connect Room, a real-time peer-to-peer communication platform for Northeastern University, integrating WebRTC for video calls, Socket.io for real-time messaging, and Node.js with Express for a scalable backend.
- Built a state-managed React frontend using Redux for efficient data handling and Material-UI for a responsive, user-friendly interface.
- Engineered a scalable backend with Node.js and Express, ensuring real-time presence tracking and seamless communication through WebSockets (Socket.io).
- Designed a MongoDB-backed data storage system, optimizing database queries for efficient retrieval and persistence of user sessions, messages, and room states.